Chapter 20

Production of Zinc, Cadmium, and Their Alloy Powders☆

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Chapter Outline

PRODUCTION OF ZINC POWDERS

There are different methods known for producing zinc powder, among them melt atomization, inert gas condensation, electrolysis of aqueous solutions, etc. Low melting and boiling points (692.5 and 1180 K, respectively) and the high chemical activity of zinc make the methods based on evaporation and condensation the most appropriate for the manufacture of zinc powder [1]. Due to a combination of high activity and high specific surface area, zinc powders find application in a variety of fields, including metallurgy, chemistry, medicine, electrical engineering, etc.

Evaporation-Condensation Method

The evaporation-condensation or physical vapor deposition (PVD) method comprises melting the zinc, evaporation of the melt, and condensation of the vapor in a neutral gas medium. In commercial production, the evaporation is carried out either in externally heated equipment or in electrothermal or induction furnaces. The continuous process of zinc powder production (Fig. 20.1) includes three sequentially arranged installations: a reverberating furnace for zinc melting and partial elimination of iron and lead, a rectification column, and a condenser.

The charge containing zinc of various grades is loaded into the gas-heated reverberating furnace that has three sections: a melting area, a liquation section, and a superheater. In the melting area, the charge is melted at about 820 K; in the liquation section, some of the iron precipitates at 740–770 K as a sediment of ferrous zinc and lead, which is periodically removed. The dross accumulating on the surface of the melt is skimmed once a day and sent for recycling in kilns or muffle furnaces. Zinc with an iron content of <0.1 wt% is fed into the superheater at a temperature of about 870 K. From the superheater, the molten zinc is fed by a chute to a rectifying column (Fig. 20.2) to one of the carborundum trays in the upper part of the column. The shaft of the column is lined with chamotte bricks and equipped with a gas supply duct and a waste gas flue. The shaft of the column consists of 20–24 W-shaped trays in the combustion section and 17–22 flat trays in the non-heated section.

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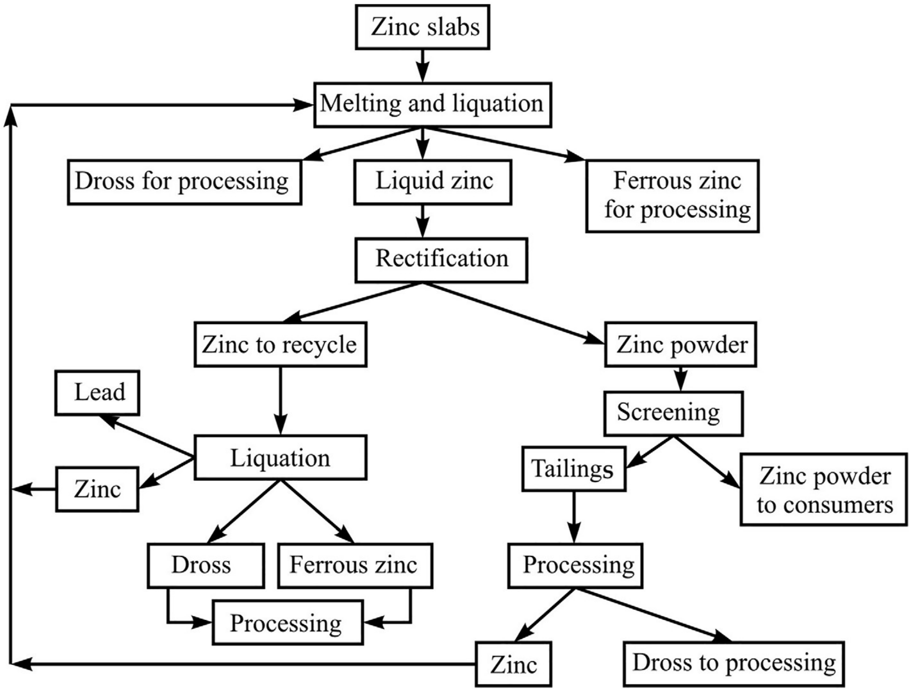
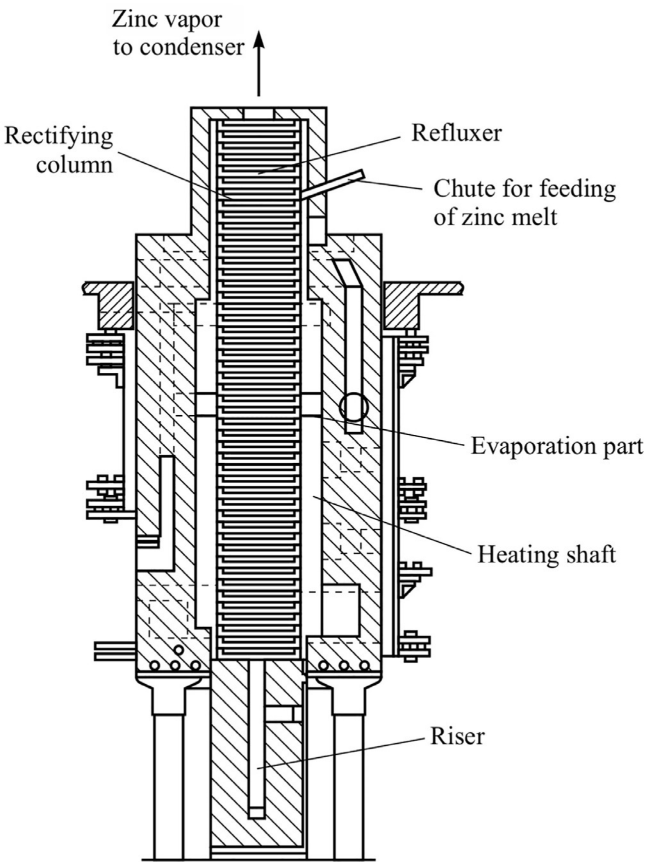


FIG. 20.1 Flow-sheet of the process of producing zinc powder by rectification of the melt.

FIG. 20.2 Rectification column.



Most of the zinc evaporates in the combustion chamber of the column at 1470–1570 K. The zinc vapor rises countercurrently to the metal melt flowing down from the top trays, gives up some heat to the melt, and reaches the condenser, where condensation of the vapor and the formation of solid particles of zinc take place at temperatures of 520–670 K. The target is a highly active polydisperse powder.

Condensation occurs according to one of two roads: vapor-liquid-crystal or vapor-crystal. The first road results in the generation of granules formative the coarse fraction. The condensation in accordance with the second road proceeds in two condensers at a significant temperature difference between the incoming gas and the condensation surface. Tables 20.1 and 20.2 present the condensation conditions and the basic parameters of the process.

The temperature within the condenser volume varies depending on the distance from the wall: 340–360 K at a distance of 25 mm; 450–500 K at 1000 mm; and 530–620 K at 1750 mm. The condensers are periodically cleaned by vibratory or mechanical methods. After leaving the condenser, zinc powder undergoes separation by sieving. The 0.3 mm tailings are recycled. The reflux part of the condensate amounting to up to 30% of the zinc charge, enriched in high-boiling components, passes from the lower trays of the column (see Fig. 20.2) through a hydroseal to a recirculating tank.

An interaction of silicon carbide and iron in the reflux flow occurs at 1370–1570 K, according to the reaction:

$$\mathrm{Fe_{Zn}} + n\mathrm{SiC} \leftrightarrow \mathrm{FeSi}_n + n\mathrm{C};\quad \Delta G^0_{1500} = -16{,}000\,\mathrm{J/mol}$$

The presence of iron in the zinc melt impairs the equipment; an increase of the iron content in the zinc from 0.04% to 0.06% reduces the service life of the column by 20%. Zinc is therefore subjected to liquation refinement in order to reduce the iron content to 0.02% or lower. Besides iron, the zinc charge usually also contains lead, cadmium, copper, and arsenic. The entire cadmium content passes to the zinc powder due to its lower boiling point. The lead content of the condensate, irrespective of its content in the zinc charge, can be reduced to a minimum under a certain temperature condition. Copper does not interact with silicon carbide. Due to its accumulation in the recycled zinc, the formation of brass occurs that is prone to precipitate on the lower trays of the column, thus hampering the process. Aluminum present in the charge in the amount of 0.01–0.05 wt% and concentrating in the reflux is useful for refining the zinc.

Zinc from the precipitation tank is fed into the reverberatory furnace. In the case where zinc oxide is to be separated, the reflux is taken out of the main cycle and processed in liquation and retort furnaces.

A process comprising evaporation of the melt with simultaneous refining can also be used. It allows the content of high boiling point metallic impurities such as iron, copper, etc., to be reduced as much as 30 times. The installation yields up to 12 t/day. Lowering the process temperature as well as using more pure source zinc also reduces the content of iron, copper, lead, tin, arsenic, and other impurities.

A technique has been developed that permits the condensation of zinc vapor to be carried out in a single water-cooled condenser. The content of the fraction <10 μm in the powder increases up to 30–45 wt%. The tightness of the equipment accounts for a lower risk of autoignition of zinc vapor in the condenser; the conditions of column maintenance are improved.

The following advantages are features of the rectification method:

1. Less pure, that is, a less expensive zinc source may be used, such as scrap.
2. Controllable powder properties.
3. Higher specific surface area of powder particles.
4. Powder particles of a regular spherical shape are produced.

Good technical performance and high economic indicators are features of the electrothermal process. The electrothermal furnace (Fig. 20.3) of $520 \times 520\,\mathrm{mm^2}$ in area is charged with blast-furnace coke sized between 380 and 20 mm. Graphite plates are connected to copper electrodes. The graphite electrode in the upper part of the furnace is sunk into the coke to a depth of 900 mm at a distance of 2550 mm from the lower graphite plates. Liquid metal is fed by metering ladles into the upper part of the furnace. The remaining metal melt is tapped through the tap hole while zinc vapor passes through a pipe connected with the condenser. The power rating of the transformer is 360 kVA.

The condensation plant (Fig. 20.4) consists of two main condensers with a surface area of 65 $\mathrm{m^2}$ and one additional condenser with a 37 $\mathrm{m^2}$ surface area. The gas from the additional condenser is directed into a cyclone cooler and further into the cyclone and a bag filter. The inert gas from the bag filter is recycled to the first condenser via the vapor line. The flow rate of the recirculating gas controls the particle size of the powder. The capacity of the blower is of 9.9 $\mathrm{m^3}$/min, pressure 17.4 kPa. The capacity of the installation amounts to 2000 t/year with power consumption 1000 kWh/t zinc powder; the consumption of coke is 2 kg/t of powder.

TABLE 20.1 Operating Modes of Rectification Columns (I, II) and Particle Size Distribution

Operating Mode	Temperature (K)			Composition of the Gas Phase in Condensers (wt%)						Particle Size Distribution (wt%)					Zinc Powder Yield (t/day)
	In Heating Shaft	In column condensers		I			II			Fraction (mm)					
		I	II	CO	CH_4	H_2	CO	CH_4	H_2	+2.5	–2.5 +0.315	–0.315 +0.16	–0.16 +0.071	–0.071	
1	1530–1540	660	630	1.9	–	–	1.7	–	–	0.8	4.2	0.1	2.1	92.8	14.1
2	1540–1545	630	600	1.7	0.32	23.6	1.4	0.33	21.0	2.4	5.2	0.3	2.4	89.7	14.7
3	1530–1540	630	640	7.0	0.78	13.5	8.0	1.2	18.0	6.9	1.5	0.3	2.7	88.6	14.5
4	1490–1530	630	690	2.3	0.32	18.0	2.0	0.33	13.0	3.9	15.2	3.2	7.1	70.6	13.8

Note: Mode 1, characteristic of normal operation of the condensers; mode 2, forced cooling of condenser walls; mode 3, supply of ~1 m^3/h carbon dioxide into condensers in counter flow to zinc vapor; mode 4, supply of water vapor into condenser (≈1 m^3/h).

TABLE 20.2 Typical Characteristic of Zinc Powders Produced by Rectification of the Melt

	Fraction Content (%)				Particle Size Analysis of the Fraction <0.071 mm						
	+0.16 mm		−0.16 mm		Content of Particles of Various Size (μm) wt%						
Mode	Zn Total	Zn Met	Zn Total	Zn Met	>15	10–15	8–10	6–8	4–6	2–4	<2
1	98.0	91.8	98.5	94.2	8.3	22.0	11.1	17.2	20.0	12.6	8.8
2	98.4	94.1	98.3	95.2	11.4	16.7	13.2	7.1	17.9	18.9	14.8
3	98.3	96.7	98.0	93.3	15.2	14.7	23.4	7.4	15.4	17.3	6.6
4	–	–	–	–	18.0	21.2	8.7	16.7	17.2	12.8	5.5

Note: See note to Table 20.1.

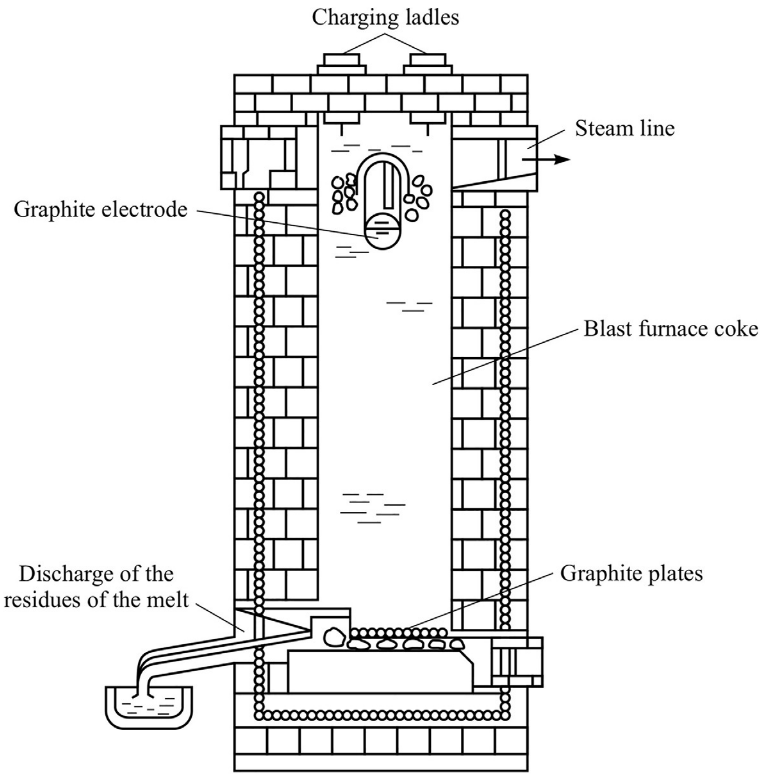


FIG. 20.3 Electrothermal furnace for production of zinc powder.

Depending on the flow rate of the recirculating gas, the powder yield amounts to 29.6%–32.4% from the first condenser, 30%–36% from the second, 12.5%–17.5% from the third, 9%–15% from the cyclone, and 5%–10% from the bag filter.

The electrothermal method does not ensure high purity of the zinc powder.

Inert Gas Condensation

A new method for producing fine zinc powder based on the inert gas condensation (IGC) process has been developed at the Institute of Metallurgy, Ural Branch of the Russian Academy of Science, and commercialized by the Fine Metal Powders Company in Yekaterinburg, Russia [2,3]. The "Tuman" ("Fog") installation of a modular type is the core of the process. The controlled process provides powders of a targeted particle size at a high yield. The total plant capacity in terms of zinc powder is up to 1500 t/year (Fig. 20.5).

The Tuman unit operates on a 12-h cycle. Chapter 5 contains more detailed information about this unit. The power consumption of the evaporator totals 46.5 kW, the consumption of the inert gas (argon) amounts to 56 m^3/month, and

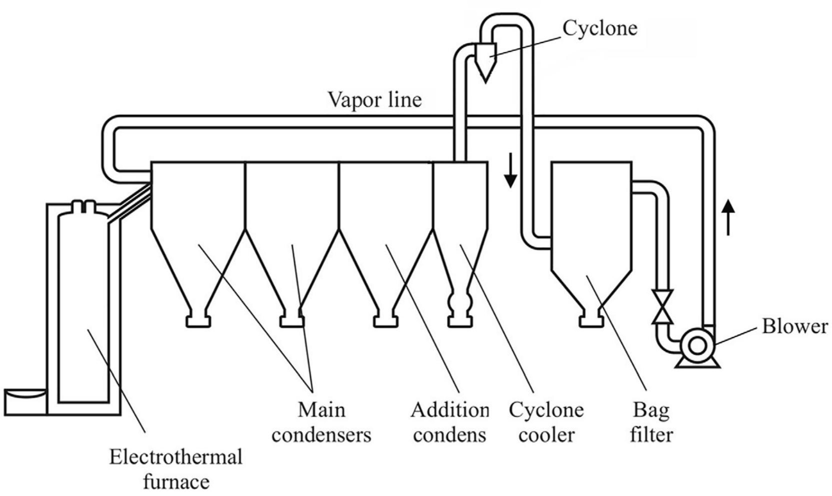


FIG. 20.4 Electrothermal process flow-sheet.

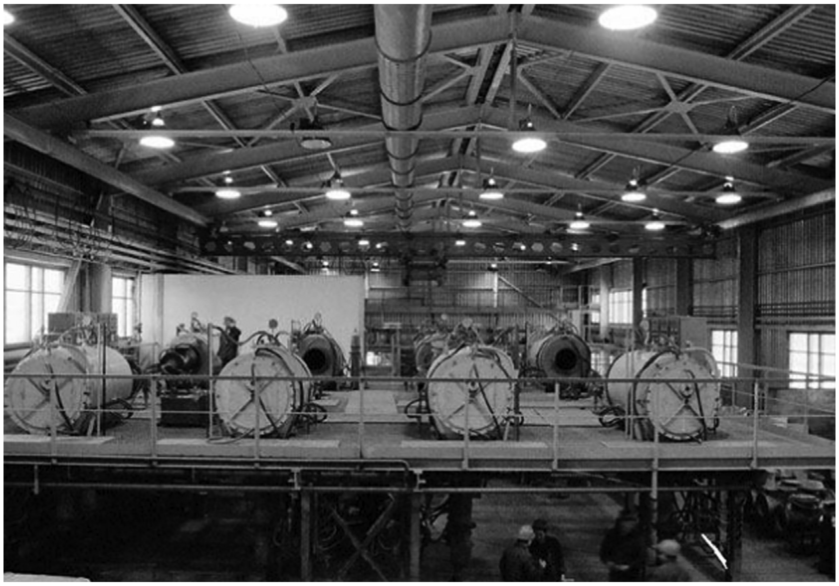


FIG. 20.5 The "Tuman" installations (the Fine Metal Powders Company's Advanced Technologies Plant).

the consumption of the cooling water is 3.5 m^3 per one modulus per hour. The zinc content is Zn 99.98% while the trace impurities of Pb, Fe, Cu, Sn, and As amount to 76, 10, 6, 1, and 2 ppm, accordingly. Spherical powder produced by the IGS process is shown in Fig. 20.6.

Melt Atomization

As shown in Fig. 20.7, zinc melt is fed into the atomization nozzle via a tube made of quartz glass, graphite, or ceramic. Gas atomization can result in elongated particles (Fig. 20.8).

A flaky powder made by centrifugal atomization is illustrated in Fig. 20.9. The superheated melt is fed through a nozzle onto a dram having a corrugated surface, with the corrugation being made parallel to the axis of dram rotation.

To obtain fine powder by gas atomization, different measures can be used such as increasing the gas pressure and, appropriately, the gas flow rate; ultrasonic impact on the gas stream; heating the gas prior to feeding it into the nozzle [4]; and nozzles of special design [5,6]. Information about hot gas atomization and different designs is in Chapter 4.

As a rule, water atomization is not used for metals with a melting temperature below 773 K because of the formation of extremely irregular particle shapes due to extremely rapid solidification. Zinc is the lowest melting metal that is water atomized commercially. Thus, zinc powder for alkaline manganese batteries is produced by melt atomization under water pressure of about 5.5 MPa. The median mass particle size of this powder amounts to 180 μm with a geometric standard deviation about 2.34 [7].

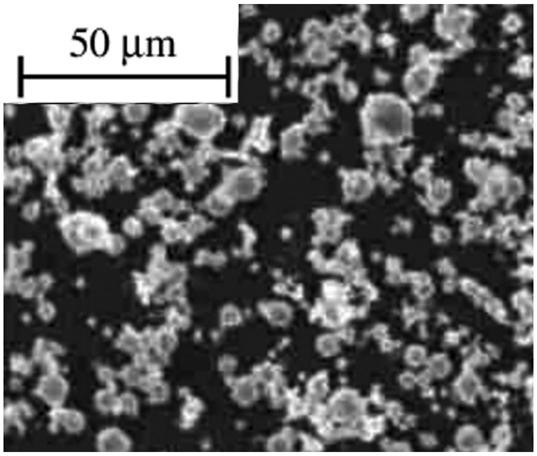


FIG. 20.6 Spherical powder produced by the ICG process.

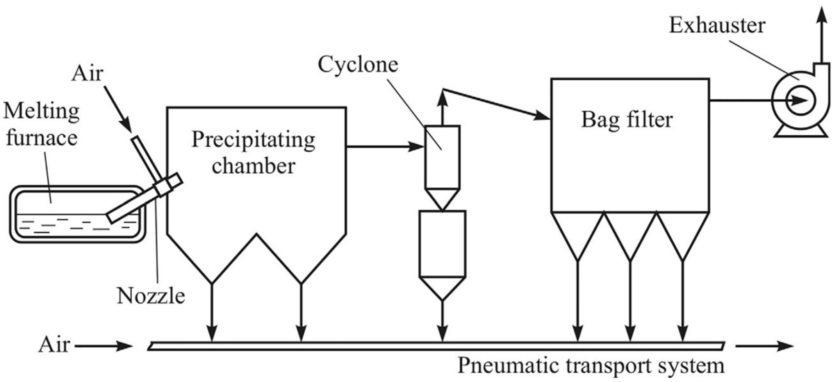


FIG. 20.7 Flow-sheet of producing zinc powder by the atomization process.

FIG. 20.8 Elongated particles of powders produced by the gas-atomization process.

Electrochemical Method

The electrochemical method allows the production of high-purity powders.

Cathodic precipitates of various types can be obtained:

- Dense layers of flaky or crystalline particles; to obtain powder, these particles must be subjected to disintegration.
- Spongy soft precipitates easy for attrition.
- Loose precipitates of fine powder that do not require additional treatment.

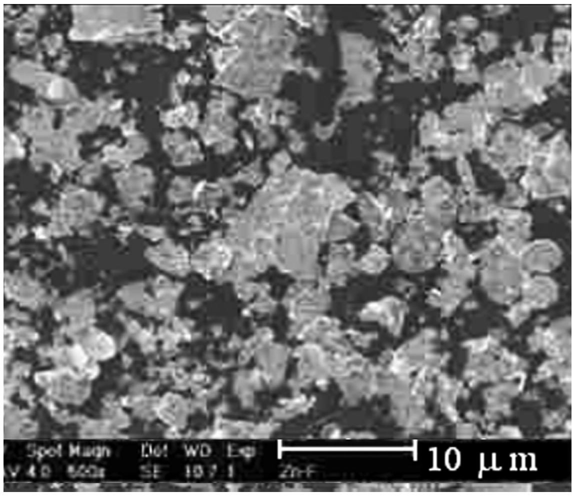


FIG. 20.9 Flaky powder produced by centrifugal atomization.

Zinc sponge is produced from alkaline zincate electrolytes (18 g/L Zn, 250 g/L NaOH) obtained either by leaching oxidized zinc-containing source material with an alkaline solution or from recycling wastes from the manufacture of benzidine, hydrosulfate, etc.

The cathodes are conventionally made of nickel, stainless steel, or magnesium. The electrolysis parameters are a voltage of 3.5–4.0 V, a current density of 2700 A/m^2, and a temperature of 293–323 K. The spongy cathodic precipitate easily slides down the cathode. After washing, it is used as a reducing agent in chemical processes. Dry powder contains 99.3% Zn [1].

A process is known that uses an electrolyte consisting of sodium hydroxide and sodium zincate, Na_2ZnO_3 [8]. A zinc content of about 55 g/L yields a deposit of the suitable particle size. The current density is about 1076 A/m^2. Under these conditions, the deposit is a loosely adherent sponge that can be easily scraped or brushed off the cathode and processed to powder.

In the production of zinc powder from industrial wastes, the latter are leached in a 6-M solution of sodium hydroxide at 330 K, bringing the zinc content of the solution to 30–40 g/L. Then the alkaline solution of the sodium zincate undergoes electrolysis at a temperature of 310–330 K and a current density of 2500–3000 A/m^2. The zinc sponge is removed by scraper blades. The precipitate is reduced to powder in an electrolyte medium by the disperser rotating at a speed of 1000–1500 rpm. The alkali residues are washed off the zinc powder.

The principal parameters of the electrochemical process are given in Table 20.3.

Dendritic deposits are formed if there are obstacles on the electrode that hinder the transfer of discharging ions to the electrode surface. They are created using various polarization methods.

The work [9] contains study results of the effect of the polarization conditions on the growth dynamics and structure of electrolytic dendritic zinc deposits obtained at room temperature from electrolyte with 0.3 mol/L zinc oxide and 4.0 mol/L of sodium hydroxide.

The following polarization modes were compared:

Galvanostatic mode by the maintenance of constant amperage of 1020 A/m^2 per the initial electrode surface.

Pulsed galvanostatic mode with congruent pulse amplitude to the amperage in the galvanostatic condition by at a ratio of pulse to pause lengths of 15/15 and 30/30 s.

Potentiostatic mode with constant polarization of –0.4 V relative to the nonpolarized zinc electrode in the solution.

Combined mode including the galvanostatic mode applied for 5 min and the pulsed potentiostatic mode with alternation of the pulse with amplitude –0.4 V and a pause in the 30/30 s ratio.

The deposit obtained in galvanostatic conditions has a typical dendritic structure (Fig. 20.10A). At high magnifications (Fig. 20.11A), it is shown that the real surface of dendrites is not flat but rather a honeycomb structure with crystallographic orientation, as in hexagonal lattices. The thickness of the cell walls is virtually constant over the whole field and is about 40–50 nm; these cells are not smooth and have a columnar structure. These honeycomb structures have extremely low packing density and high specific surface area.

During pulsed galvanostatic electrolysis, the concentration of discharging ions in the diffusion layer increases during pauses between pulses. When current is recurrently switched on, the overvoltage is not as high as it was at the beginning of the electrolysis because of the increase in the near-electrode ion concentration and deposit surface area due to the previous

TABLE 20.3 Specification of Electrolytic Methods of Manufacturing Zinc Powder

Process	Electrolyte Composition (kg/m^3)	Current Density ($\kappa A/m^2$)	Temperature (K)	Process Characteristics
AS	$ZnCl_2$ of 15–80; pH = 4.4–5.8	0.1–0.8	293–363	Graphite anode; electrolyte is periodically adjusted by concentrated solution of $ZnCl_2$
AS	$ZnCl_2$ of 20–40; NaCl of 300	0.54	–	Current efficiency of 95%; chlorides are easily washed out of the sponge
AS	Zn of 15–22; NH_4NO_3 of 10; HNO_3 to 63	0.2–0.3	293	…
AS	Zn of 10; NaOH of 200	1.1–1.5	303	Nickel anode, magnesium cathode; spongy deposit; powder yield of 90%
AS	Zn of 8–16; NaOH of 100–200	0.5–1.0	293–323	Current efficiency of 75%–90%; deposit build-up time of 2–4 h; active zinc content of dry powder of 90%–97%
AS	$ZnSO_4$ of 20–40; NaOH of 230	0.5–2.5	298–318	Powder apparent density of 0.32–0.45 g/cm^3; specific surface area of 0.35–0.58 m^2/g
AS	$ZnSO_4$ of 0.5–200; NaCl of 20	1.8–20.8	308–343	–
RSPh	Na_2CO_3 of 159; ZnO	1.0	293	Raw material is ZnO; metallic zinc content of product of 90%; drying temperature of 333–338 K; particles are small, porous
AS	Zn of 40; NaOH of 220; oleate Na of 0.05; $CuSO_4$ of 0.5	3.0	333	Powder obtained in potentiostatic conditions ($E = -2.04$ V, normal hydrogen electrode) features elevated activity in the synthesis of rongalite
AS	$ZnSO_4$ of 60–65; NH_4Cl of 150; H_3BO_3 of 20; NH_3 of 20–25	0.3	290–293	Sedimentation of the powder at a potential of (−1.15 to −1.2) V, normal hydrogen electrode; pH = 10
RSPh	Na_2CO_3 of 159; ZnO	1.0	293	Raw material is ZnO; metallic zinc content of product of 90%; drying after washing and stabilization at 333–338 K; particles are porous

AS is aqueous solution; RSPh is the reduction from the solid phase.

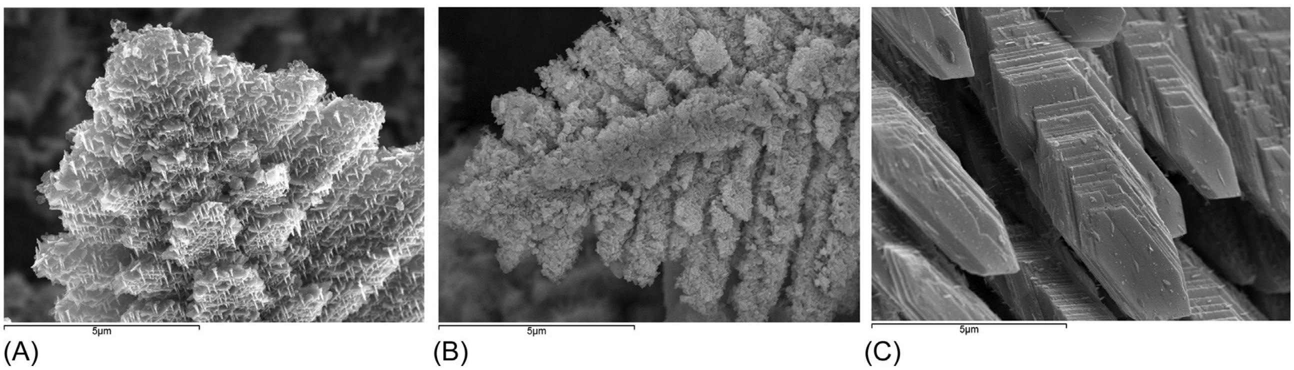


FIG. 20.10 Morphology of zinc dendritic depositions occurs in the following conditions: (A) galvanostatic, (B) combined, and (C) impulsive galvanostatic by impulse/pause equal to (15 s)/(15 s). *(Courtesy of Prof. T. Ostanina.)*

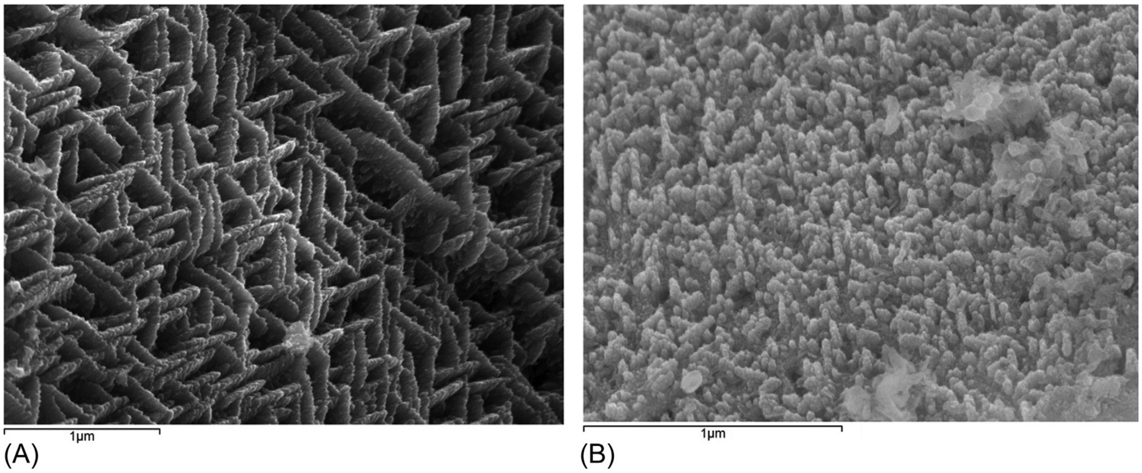


FIG. 20.11 SEM micrographs of zinc dendritic depositors occur in galvanostatic (A) and combined (B) conditions at high magnifications. *(Courtesy of Prof. T. Ostanina.)*

pulses. Therefore, the adatoms do not form new nuclei but are built into existing growth steps. In this case, a dendrite branch is of lamellar structure, in contrast to the honeycomb structure in the previous mode (Fig. 20.10C). This leads to a dramatic increase in the differential density of the deposit and a decrease in the specific surface area.

In the combined mode, there were alternating periods of growth and dissolution of the metal. A honeycomb structure forms at the stage of growth, and selective dissolution of the walls occurs during pauses. During the next potentiostatic pulse, the overvoltage remains high despite the considerable development of the surface, and the oversaturation of adatoms is sufficient for their crystallization on highly curved sites. This leads to the formation of a one-dimensional structure consisting of thin columns (with a diameter smaller than the thickness of the cell walls formed during the first stage) and measuring about 20–30 nm (Figs 20.10B and 20.11B). In the end, the volume is filled with interconnected columns forming a "vaulted" structure with an even lower filling factor than in the honeycomb structure.

The characteristics of zinc dendritic deposition at the end of the electrolysis cycle by various polarization modes are summarized in Table 20.4.

Other Methods

Mechanical grinding in various devices of ball, hammer, inertial, vibratory, jet type, etc., is one of the methods of obtaining fine zinc powder.

Air impact mills provide powders with a particle size in the range of 10–150 μm. The stability of the operation parameters and continuous recycled process are features of these plants.

TABLE 20.4 Characteristics of Zinc Dendritic Deposition at the End of the Electrolysis Cycle by Various Polarization Modes (Mean Value)

	Characteristic of Deposits			
Deposition Mode	y (cm^2)	S_m (cm^2/g)	ρ_{dep} (g/cm^3)	P (%)
Galvanostatic	0.11	302	0.18	97
Pulsed galvanostatic				
15/15 s	0.09	197	0.24	88
30/30 s	0.10	207	0.21	97
Potentiostatic	0.23	493	0.15	98
Combined	0.24	403	0.14	98

y is the length of dendrite; S_m is the specific surface of the deposition; ρ_{dep} is the deposition density; P is the porosity of the deposition; $P = 100 - \frac{100 m_{dep}}{V_{dep}\rho_{Zn}}$; m_{dep} is the mass of deposition; V_{dep} is the volume of the deposition; ρ_{Zn} is the zinc density.

Powders $<1\,\mu m$ in size can be produced by wet grinding in attritors in the presence of substances preventing auto-ignition of the powder such as nitrobenzene and salts of oleic or stearic acid. The addition of surfactants to the dispersing liquid will aid in reducing the particle size further.

Cryogenic methods enable fine and ultrafine powders to be produced. To obtain ultrafine powder, the powder is suspended in an aqueous medium and subsequently frozen under ultrasonic impact. The water is evaporated in a vacuum; the resulting product is a powder with a particle size $<1\,\mu m$.

Zinc powder can be produced as a by-product by treatment of zinc-containing wastes. The surface of the zinc-containing melt is blown with an inert gas. Zinc vapor is condensed as a fine powder on the surface of a rotating water-cooled drum (a rotational speed of 0.5–10 rpm) located at a distance of 500–3000 mm from the surface of the melt. The average size of the resulting particles is $<3\,\mu m$.

PRODUCTION OF CADMIUM POWDERS

All methods used for producing zinc powder are applicable to producing cadmium powder. Electrolytic deposition is the most common method (Table 20.5).

Cadmium powder can be obtained by cementation from a solution of copper-cadmium cake in waste electrolyte and the removal of copper from it. The composition of the solution is (kg/m^3): Cd 6–12; Ni 0.01–0.04; Fe 0.02–0.06; Cu $(0.2–0.3) \times 10^{-3}$; Co 0.08–0.12. Deposition is carried out in centrifugal reactor-separators using zinc dust. The thallium presence in the solution is precipitated together with cadmium but, after the bulk of the zinc dust is utilized, thallium goes back into the solution. The cementation of cadmium is finished when the concentration of thallium in the solution gets close to the initial level. Ions of silver and indium promote the cementation of cadmium. The duration of the operation is 35–40 min. The precipitate contains up to 95%–99% Cd. Power consumption is 1–2 kWh per solution volume of 1 m^3.

Cadmium powder with high electrochemical activity consisting of structurally uniform dendrite particles can be obtained from aqueous solutions on a horizontal cathode at varying current density. The composition of the electrolyte is (kg/m^3): Cd 5.6–20.2; acetic acid 90; potassium acetate 100–150; diethylenetriamine-penta-acetic acid 7.8–28 or diamine-cyclohexane-tetra-acetic acid 8.6–31. The current density rises with the duration of the process from a minimum value ($i_0 = 1.0–1.5\,\kappa A/m^2$) to a maximum of $i_\tau = 3.0\,\kappa A/m^2$. The duration of growth of the disperse layer approaches 0.3–0.5 h. The powder is washed with a 5% solution of acetic or sulfuric acid, deionized water, or acetone and dried in a vacuum. The cadmium anode without filler has a porosity of 60%–70% and ensures a discharge current efficiency of 90%.

PRODUCTION OF CADMIUM AND ZINC ALLOY POWDERS

Zinc- or cadmium-based alloys are conventionally produced by the electrochemical method (see Table 20.3) or by atomization (Table 20.6).

For producing an alloy with an atomic fraction of zinc of up to 40%, solutions of zinc and cadmium sulfate are used. Powders of targeted composition have been obtained by varying the composition of the solution and the current densities (Table 20.7). Alloys are precipitated at galvanostatic conditions.

The alloy powder corresponds to a conglomerate of particles of dendrite shape with their longitudinal dimension approaching twice the cross-sectional dimension. An increase in the current density from 1000 to 15,000 A/m^2 leads to a reduction in particle size by a factor of 1.5–2.0 (from $250 \times 140\,\mu m$ to $150 \times 75\,\mu m$). A change in the composition of the solution does not affect the particle size significantly.

Zinc-copper alloy powders can be produced by electrolytic deposition from cyanide as well as from pyrophosphate, ammoniate, tartrate, citrate, and other solutions. For some alloys (Zn-Fe, Zn-Sn), there exists a nearly direct relation between the concentration of the metal in the solution and the ratio of the atomic fractions of the corresponding metals in the alloy. The composition of the alloy also depends on the current density, the temperature of the electrolyte, and the intensity of agitation.

Spherically shaped powders of zinc, cadmium, and their alloys can be produced by atomization of the melt that contains additives of alkali metals: Li, Na $\geq 0.001\%$ or K, Ce, Rb $\geq 0.01\%$. Atomization of a zinc melt containing 0.4% Na (the temperature of the melt is 770 K, air temperature is 320 K, and pressure is 0.3 MPa) provides a powder with up to 80% particles of spherical shape in the size range of 150–480 µm. The apparent density of the powder is 4.2 g/cm^3. Without the addition of sodium, particles obtained under the same conditions have an irregular shape, the metallic luster is less marked, and the apparent density is 3.2 g/cm^3.

TABLE 20.5 Specification of Electrochemical Methods of Manufacturing Powders of Cadmium and Its Alloys

Process	Electrolyte Formulation (kg/m^3)	Current Density ($кA/m^2$)	Temperature (К)	Process Characteristics
Cadmium				
EAS	$CdCl_2$; HCl	30–70	300	Sponge of fine two-dimensional dendrites; cadmium in the form of chloride
EAS	Cd of 0.11–11; H_2SO_4 of 50	0.05–0.2	300	Constant overvoltage of 80–150 mV; particle shapes dendrites and needles with a length in the range of 70–300 μm
EAS	$CdCl_3$	3–0.75	298	Replacement of cementation by electrolysis reduced cost due to dust retention by a factor of 5
EAS	$CdSO_4$	5–15	298	Current efficiency of 86%–93%; particle size of 50–80 μm; chemical cell capacity increases by 20%–30%, and their service life increases by a factor of 1.5–1.6
EAS	Cd_{start} of 100;Cd_{end} of 20; H_2SO_4 of 10	0.6	298–301	Zinc present in the solution to 30–40 kg/m^3 is admissible; current efficiency of 93%–96%; deposit building up time of 24 h
ENAS	Dimethylformamide-solvent; NH_4BF_4 of 210; $Cd(BF_4)_2$ of 28.6–114.4	4	298	Constant overvoltage of 0.1–0.25 V; rotating electrode; deposit is washed off; at a rotational speed of 20–270 s^{-1} and operating time of 60 min; average particle size is of 6–29 μm
RSPh	CdO in form of tablets, NaOH of 40	3–4	298	Raw material is CdO, nickel anode; drying for 90 min at 323 К; current efficiency of 96%; particles: dendrite, polydisperse, porous
Cadmium-zinc alloy				
AS	$CdSO_4$ of 8.4–37.5; $ZnSO_4$ of 2.4–48.3	1–15	293	Current efficiency of 90%–96%; powder composition depends on deposit build-up time and concentration of cadmium in the solution
Cadmium-iron alloy				
AS	Cd + Fe of 10; (NH_4) $2SO_4$ of 50; pH 2.7	35	293	Independent regulation of anodic currents of cadmium and iron; current efficiency of 65%–67%; average particle size of 5 μm
Cadmium-nickel alloy				
AS	($CdSO_4$ + $NiSO_4$) 0.2 mol/L; NH_4Cl of 5.4	3	293	Current efficiency of 65%; average particle size of 2 μm; phase composition-mechanical mixture of cadmium and nickel: (Cd/Ni) powder = (Cd/Ni) solution

EAS is electrolysis of aqueous solution; ENAS is electrolysis of nonaqueous solution; RSPh is the reduction from the solid phase; AS is aqueous solution.

A technique is known for producing zinc alloy powders by adding 5%–80% iron and 0.2%–5% aluminum or copper to the molten zinc. The melt is cooled and ground. The powder has elevated corrosion resistance and is hardly oxidized, owing to the protective cathodic action of iron. It is used as filler in resins.

SAFETY MEASURES IN ZINC AND CADMIUM POWDER PRODUCTION AND HANDLING

The high chemical activity of fine and, particularly, ultrafine, zinc, and cadmium powders accounts for their flammability and explosiveness. Due to this fact, fire and explosion safety measures are necessary in zinc and cadmium powder

TABLE 20.6 Typical Characteristics of Zinc Powder Produced by Ultrasonic Atomization of the Melt

Medium	Pressure (MPa)	Fraction Yield (%) Particle Size (μm)								Metal Consumption (kg/min)	Content of Oxygen in Fraction <63 μm (wt%)	Apparent Density (g/cm³)
		+120	–120 +80	+100	–100 +50	–80 +63	–63 +40	–50	–40			
Argon	–	10.6	20.3	–	–	10.4	51.1	–	7.6	–	0.02	–
Air	–	25.5	29.1	–	–	13.8	27.9	–	3.7	–	0.14	–
Vacuum	0.1	–	–	69.5	18.5	–	–	12	–	10.2	–	3.6
Vacuum	0.2	–	–	7.0	8.0	–	–	85	–	35.0	–	3.9

TABLE 20.7 Effect of Electrolyte Composition on Powder Composition and Cathode Potential (Current Density of 15,000 A/m^2)

Content in Solution (g-ion/L)		Atomic Fraction of Zn in Powder (%)		
Cd^{2+}	Zn^{2+}	Actual	Estimated	Cathode Potential (V)
0.18	0.15	6.9	6.5	–0.77
0.18	0.075	6.5	7.5	–0.78
0.18	0.03	5.6	6.2	–80
0.09	0.15	14.5	15.0	–78
0.07	0.15	20.6	22.3	–0.79
0.045	0.15	29.5	28.6	–0.81

production and handling. For powder with particles <74 m, the minimum spontaneous ignition temperature is 870 and 843 K and the minimum ignition energy is 640 and 4000 mJ for zinc and cadmium powders, respectively. The zinc sponge is removed by scraper blades. The lower explosive limit for zinc powder is 480 g/m^3 (see Chapter 27). The spontaneous ignition temperature depends on the chloride content (>0.05%); moisture stimulates the spontaneous ignition process.

The presence of zinc oxide determines the degree of toxicity of zinc powder. Inhalation or swallowing of zinc powder injures breathing organs and causes gastrointestinal disturbances. Under the workplace exposure limits, in the European PM industry the long reference period exposure limits (WELs) for zinc respirable dust is 4 mg/m^3. The threshold limit value (TLS) in the air at the workplace (in terms of zinc oxide) adopted by CIS 12.1.005–88 is 0.5 and 0.05 mg/m^3 [10]. The TLS in drinking water (in terms of metal zinc) is 1.0 mg/L [11]. Techniques for the prevention of exposure to hazards are described in Chapter 27.

APPLICATIONS

There are two basic sectors of application of zinc powders:

- Protective coatings, including zinc-rich primers and paints; powder coating; and sherardizing.
- Chemical processes: purification of zinc solutions before electrolysis at electrolytic zinc plants; manufacture of reducing agents (sodium and zinc hydrosulfite) for the paper and textile industries; in the mining industry (for the extraction of gold, silver, and other precious metals); and manufacture of alkaline batteries.

For manufacturing alkaline batteries and for other chemical applications, high-purity zinc powder is needed, which is produced using extra pure special high grade (SHG) zinc.

Zinc Powders in Protective Zinc-Rich Coatings

Zinc-rich coatings combine the advantages of metallic (cathodic protection) and paint (barrier, hydro insulating effect) coatings that make them perfectly suitable for "cold" galvanizing of metals.

The most important applications of zinc-rich primers and paints are coatings exposed to severe corrosion-hazardous conditions: offshore oil fields, the petroleum industry, power engineering, sea container production, shipbuilding, and ship repair. These materials are also widely used for protecting metal structures in the construction of buildings, bridges, etc.

Zinc-containing protective materials are divided into zinc primers containing 25%–75% zinc and zinc-rich primers (85%–92% Zn). They are also distinguished by the type of binding agent: (a) organic compounds (e.g., systems based on epoxy ethers and epoxy polyamides or moisture-curing systems based on polyurethane, vinyl, and vinyl chloride rubbers); and (b) inorganic compounds on the base of ethyl silicates and alkaline silicates. Inorganic compounds display higher density and greater abrasion resistance, but require a more careful pretreatment of the surface and perform lower flexural strength than organic zinc-rich compounds [12].

Zinc powders used as pigments in anticorrosive coatings should meet certain requirements. Thus, the metal zinc content in such powders should not be lower than 95% with a minimum of zinc oxide present. Limits are set on the content of lead (<0.15%) and iron (<0.004%); no moisture is admissible. Lead impairs the quality of the coating and threatens labor safety conditions while welding. Elevated iron content of the powder reduces the protective potential of the primer while moisture causes caking of the stored powder and creates a risk of emission of hydrogen prone to destroy the protective effect of the coating [13,14].

Zinc powders conventionally used in zinc-rich protective coatings have their particle size in the range of 3–12 μm (Table 20.8). The particle size of powders influences the density of particle packing. The maximum packing density of about 61% of the solids content (by volume) is peculiar to powders with an average particle size of 6 μm.

In recent years, powders with flaky particles (see Fig. 20.9), 0.2–0.4 μm thick and 20–30 μm long, have been used in primers and paints. When flaky powders are used, reliable corrosion protection is achieved at a powder content of 80% of the dry coating that reduces the consumption of zinc and increases the covering capacity and the elasticity of the coating.

TABLE 20.8 Specification of Zinc Powders for Protective Coatings

Characteristics	ASTM Standard for Zinc Dust Type III (For Use in Zinc Rich Coatings)	Manufacturer, Grade, Manufacturing Technique			
		Umicore, Belgium. Zinc Dust Grade 4P16, (Distillation)[a]	Trident Alloy, United Kingdom Delaville Standard CP75 Zinc Dust (Electrothermal Furnace Technology)[b]	US Zinc, United States Grade USZ # 1 XL (Distillation)[c]	Fine Metal Powders Company, Russia Zinc Powder Grade PZHD-1 (IGC Technique)
Total zinc, calculated as Zn, min.%	99.0	99.0	98.0	98.5	99.0
Metallic zinc, min.%	96.0	96.5	95.0	95	96.0
Lead, calculated as Pb, max.%, s	0.002 (20 ppm)	0.003	0.1	0.01	0.013
Cadmium, calculated as Cd, max.%	0.001 (10 ppm)	0.0005	0.005	0.01	0.004
Iron, calculated as Fe, max.%	0.002	0.002	0.003	0.02	0.003
Medium particle size, μm	–	3.4–3.9	–	5–8	4–12
Coarse particles μm max.%					
+150	0.1	0.00	0.01	Trace	0.00
+75	0.8	0.00	0.3	0.1	0.8
+45	3.0	0.7	5.0	4.0	3.0
Specific weight, g/cm^3	–	–	7.1	7.1	7.1

[a]*Source:* http://www.jamesmbrown.co.uk/zincdust/zincdust.htm.
[b]*Source:* http://www.tridentalloys.com/Pigments.
[c]*Source:* http://www.uszinc.com/ZincDust/ProductSpecs/XL_.

Anticorrosive Zinc-Rich Coatings for Roll and Sheet Steel and Metal Wear

Protective coatings for roll and sheet steel based on pure fine zinc powder have been developed. These are DACROMET, ZINCOMET, GEOMET, PLAS, etc.

DACROMET is an inorganic coating composed of zinc and aluminum flake in a binding matrix of chromium oxides. Zinc and aluminum platelets align in multiple layers forming a metallic silver gray coating. Applied as a liquid material, the coating becomes totally inorganic after curing at 594 K. The corrosion resistance performances of DACROMET are particularly high at a low thickness (from 5 to 10 μm). DACROMET can serve as a base for most paints, including electrodeposited paint. Two DACROMET coating types are commonly used to meet the automobile industry's requirements:

Grade A: coating weight $> 24\,g/m^2$, average thickness of 5–7 μm.
Grade B: coating weight $> 36\,g/m^2$, average thickness 8–10 μm.

Coatings of greater thickness may be applied to increase the duration of the corrosion protection. Several application techniques are used industrially, among them immersion and pneumatic or electrostatic spray.

Sherardizing (Thermodiffusion Galvanizing)

Sherardizing is a very reliable and versatile method of coating steel surfaces and therefore giving high protection against corrosion and abrasion. The metal articles are heated in the presence of zinc powder (dust). The process is normally carried out in a slowly rotating closed container at temperatures ranging from 593 to 773 K (Fig. 20.12). Several aftertreatments may be applied (chrome passivation, lubrication, etc.).

A sherardized coating forms two layers of zinc-iron alloy: a diffused gamma layer containing 21%–28% of iron and a compact delta layer containing 8%–10% of iron. The thickness of the coating may be in the range of 10–110 μm.

The main benefits of the sherardized coating are the thickness of the coating remains regular and uniform, even on irregular or complex shaped and recessed components; high adhesion due to the zinc-iron alloy diffusion; no hydrogen embrittlement; versatility; easy operation; and environmental friendliness.

Usage: anticorrosion treatment of metal articles (nails, bolts, screws, cast iron articles, metal pegs, medium-sized metal objects of irregular form, etc.).

An important feature of thermodiffusion galvanizing is that it does not impair the mechanical properties of heavy-duty fixing parts (bolts and nuts).

Mechanical Plating

Steel articles are placed in a drum with a mixture of a zinc powder, an activating solution, and glass beads. The drum is then tumbled and the beads impact zinc onto the surface of the steel. The thickness of the resulting zinc coating is 10–12 μm. Also, cadmium powders and mixtures of Zn and Cd powders may be used.

FIG. 20.12 An installation for thermodiffusion galvanizing (sherardizing): general view while container is being charged.

TABLE 20.9 Powder Characteristics and Light Element Content

Zn Powder	Undersize (μm)			Hall Funnel Flow (s)	Content of Impurities (ppm)		
	δ_{10}	δ_{50}	δ_{90}		Oxygen	Nitrogen	Hydrogen
Powder 1	10	29	58	No flow	2585 ± 71	17 ± 2	23 ± 1
Powder 2	26	39	60	20.98	2551 ± 81	17 ± 2	7.6 ± 0.8

Recycling of Tungsten Carbide Powders

The most important process for hardmetal scrap direct conversion into graded powder ready for pressing and sintering is the so-called "Zn process" [15–17]. The zinc process steps are as follows: (1) Sorted, cleaned, and crashed hardmetal scrap**;** (2) Reaction in molten zinc at 1173–1273 K in a protective atmosphere (Ar/H_2); (3) Vacuum distillation at 1273–1323 K; (4) Crushing, ball milling, and screening; (5) <75 μm powder ball mill, screen, blend, carbon adjust; and (6) Graded hard metal powder for use.

Implant Production

There is a long-term interest in biodegradable load-bearing implants in medical research in general, and thus also in medical laser-bed powder-bed fusion (PBF-L) research. The third group of alloys after magnesium and iron alloy groups to be considered promising for biodegradable implant production are Zn alloys [18–20]. The first peer-reviewed publication on this subject appeared in 2007 [21].

The main objective of the work [22] was the study and development of the direct metal printing (DMP) process for pure Zn. DMP is a commercial name that refers to the same basic process in which a laser is used to selectively fuse powder along a trajectory based on a computer-aided design (CAD) file. Two different Zn powders were used for this research. The first, P1, was produced by air atomization. The second, P2, was produced by nitrogen atomization. As Table 20.9 shows, P2 has a better flowability than P1. Both powders have some porosity and the amount of pores is similar for both powders.

This study has shown that it is possible to produce high-quality Zn parts by direct metal printing (DMP). Single track scans were used to estimate suitable process parameters. In the two-dimensional experiments, the hatch spacing was chosen to balance track overlap and smoke formation. The quality of the layers after DMP largely depended on the flowability of the powder. A relative density of >99.70% was obtained for a $10 \times 10 \times 10\,mm^3$ density cube. A vertical cross-section of this sample showed that elongated grains grow in the building direction across multiple layers.

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